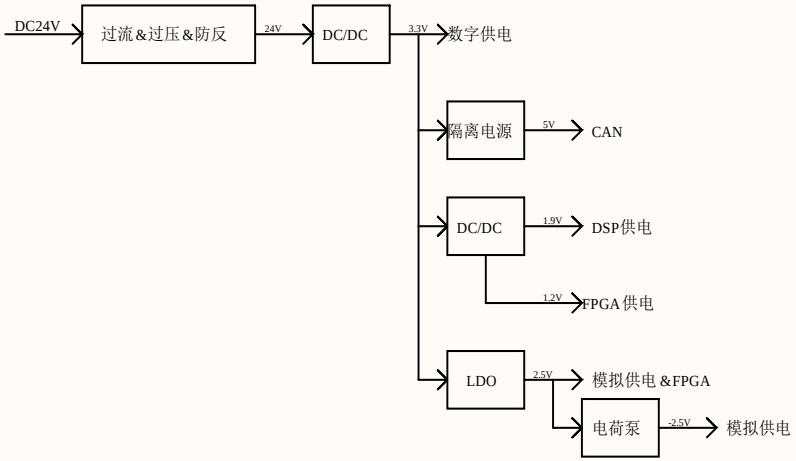
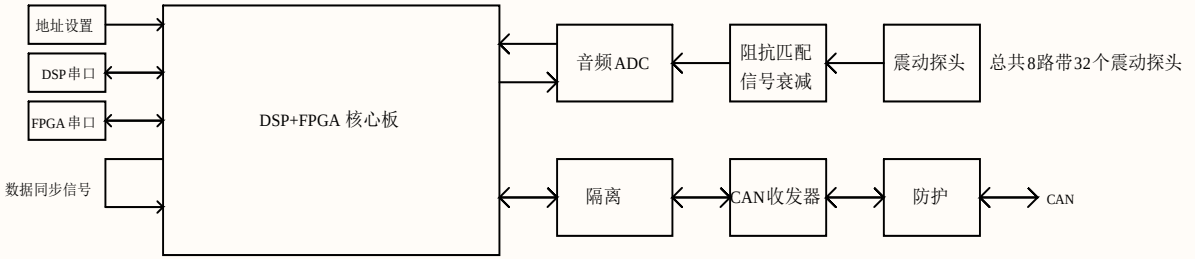
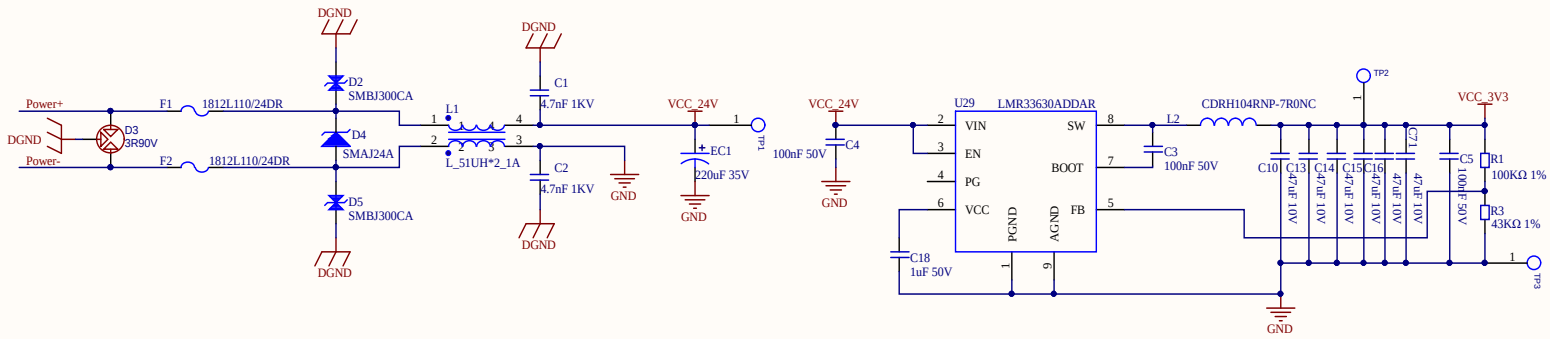
版本号	日期	修改人	修改内容
V1.0	2022.11.1	桑牧丹	新建
V1.1	2022.11.24	桑牧丹	1、添加到32路 2、传感器改为震动直采 3、更改采集相应电路
V1.2	2023.1.9	桑牧丹	1、调整外形 2、更换电源方案 3、删除FPGA复位 4、更换传感器插座
V2.0	2024.1.26	桑牧丹	1、方案整合到一张板子上 2、优化布线 3、优化电源 4、更换对外插座，输入改为单端输入

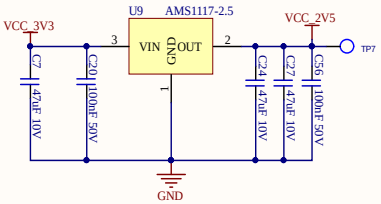
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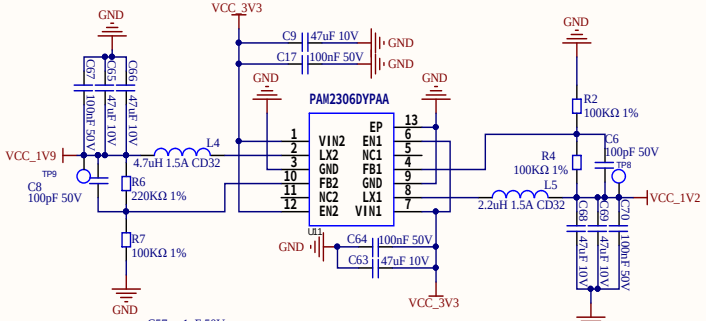


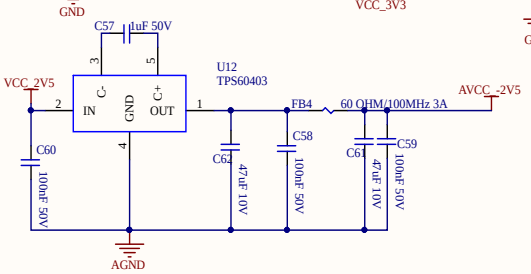


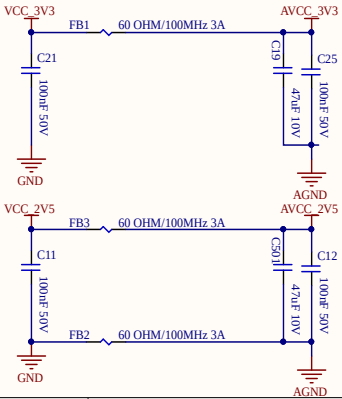
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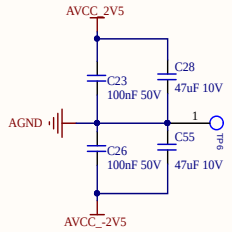


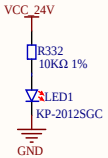


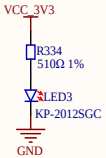


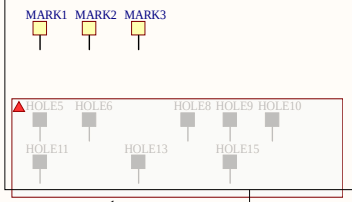




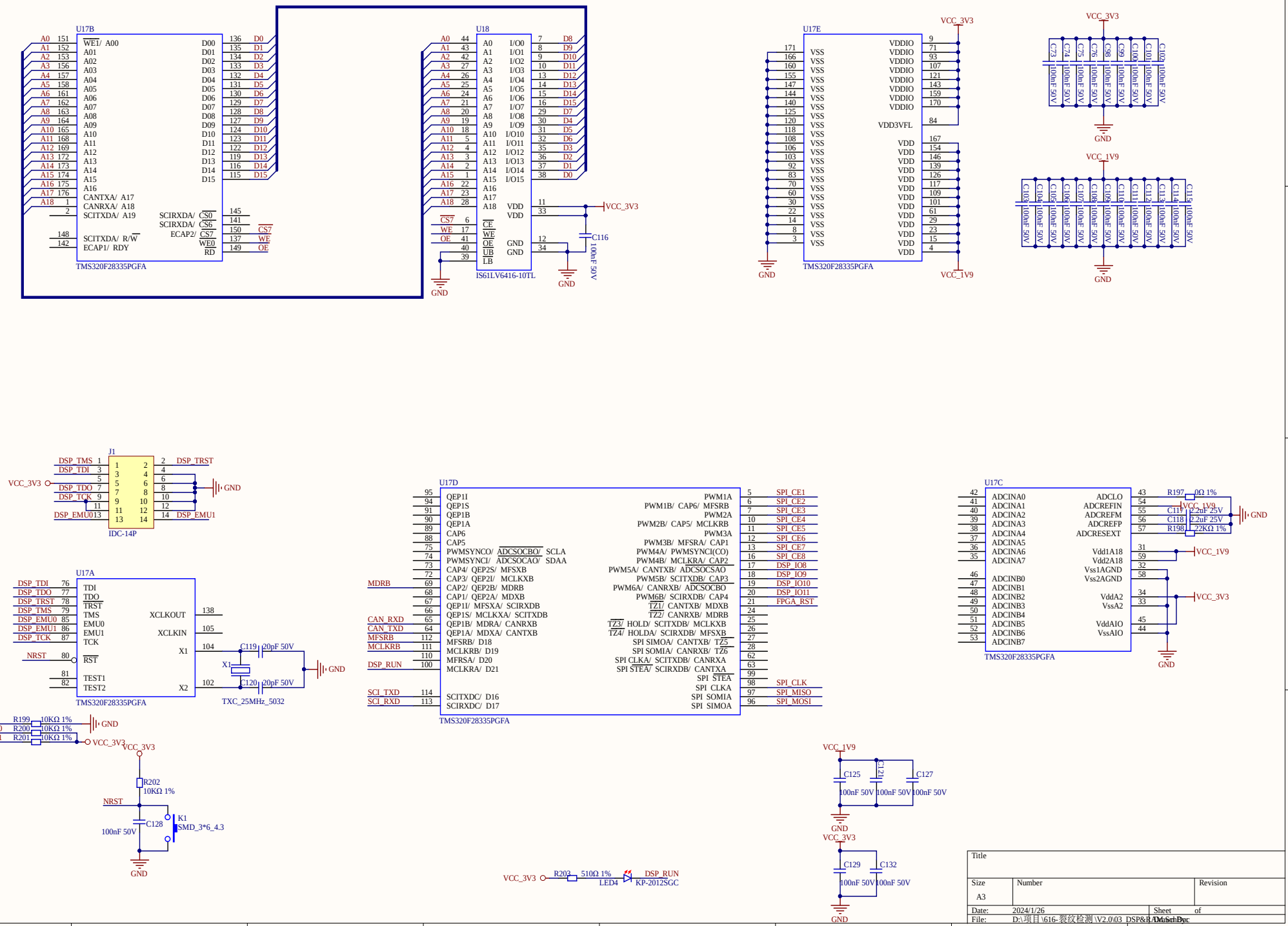


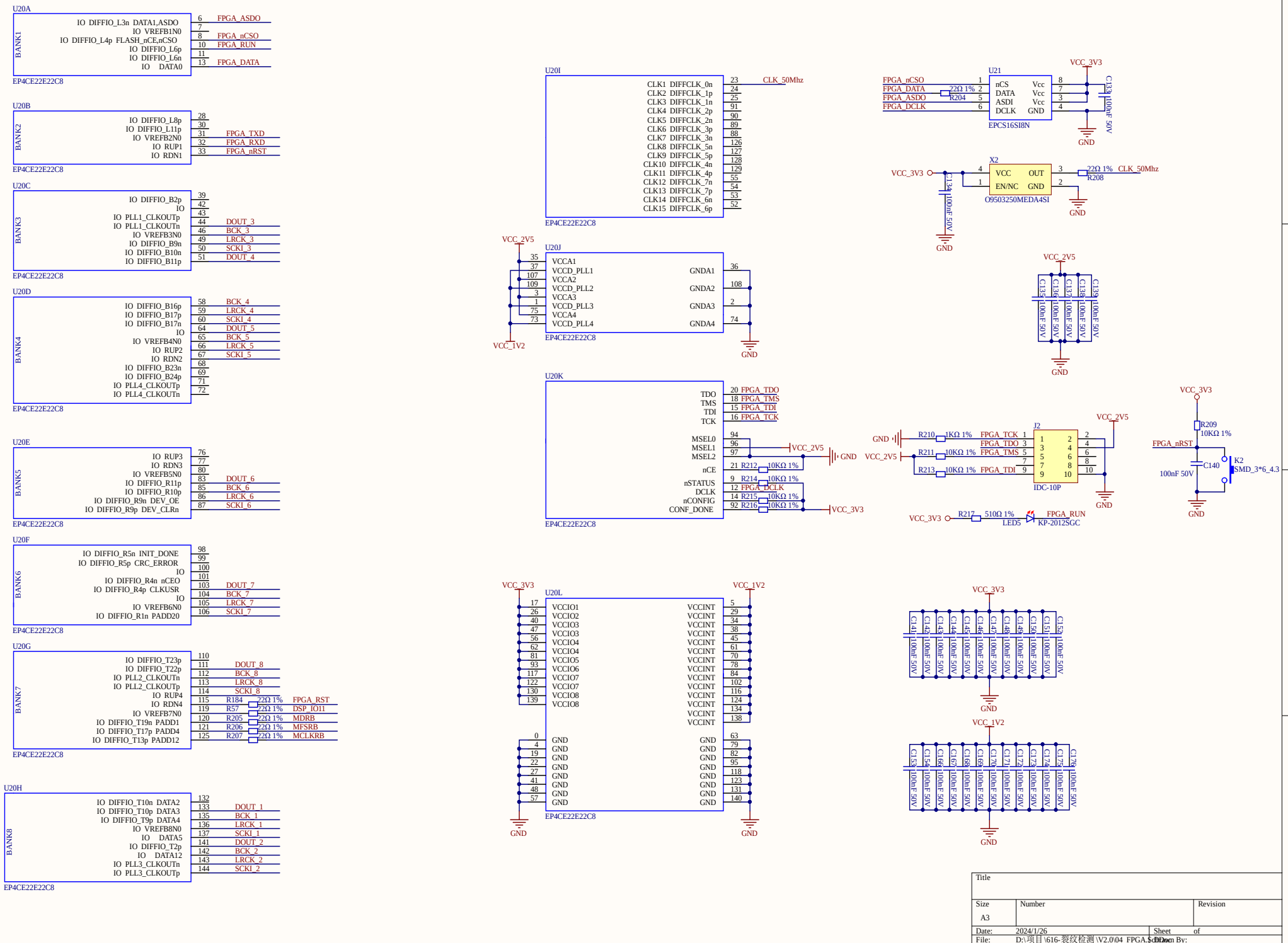


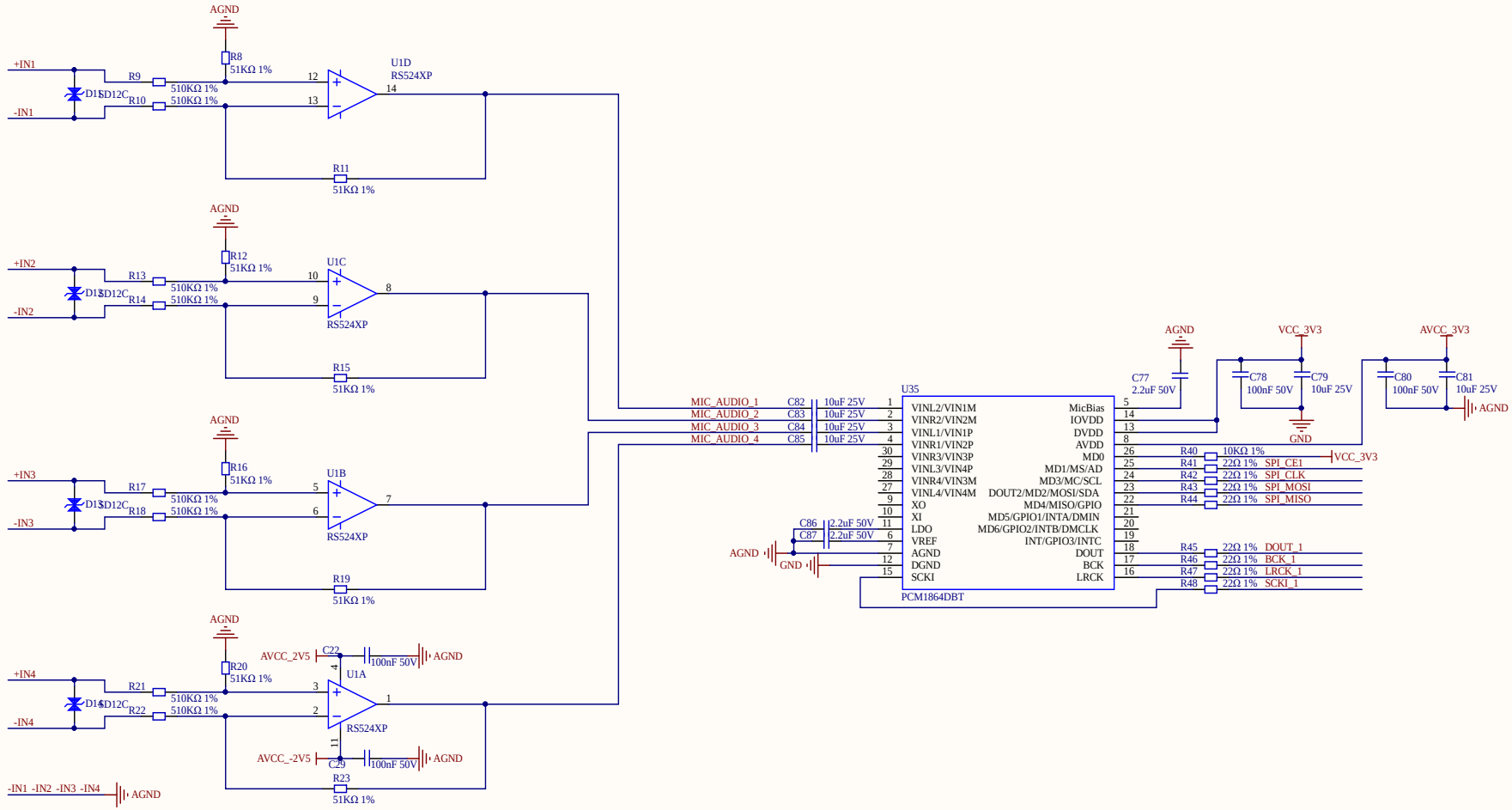




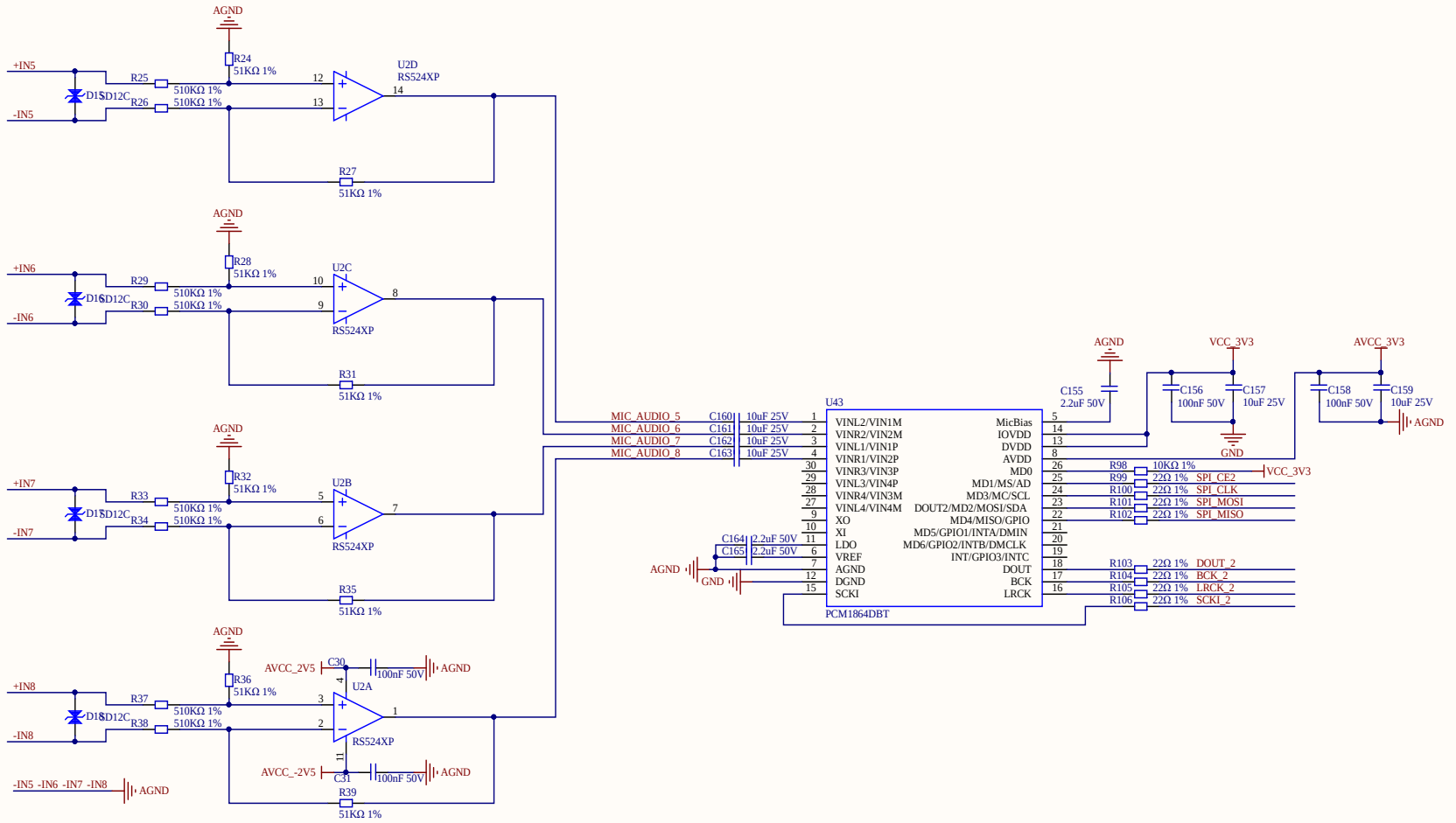
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U17B
TMS320F28335PGFA
U18
IS61LV6416-10TL
U17E
TMS320F28335PGFA
VCC_3V3
VCC_1V9
GND
J1
IDC-14P
U17A
TMS320F28335PGFA
TXC_25MHz_5032
U17D
TMS320F28335PGFA
U17C
TMS320F28335PGFA
R202
10KΩ 1%
K1
SMD_3*6_4.3
LED4
KP-2012SGC
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Size
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Date: 2024/1/26
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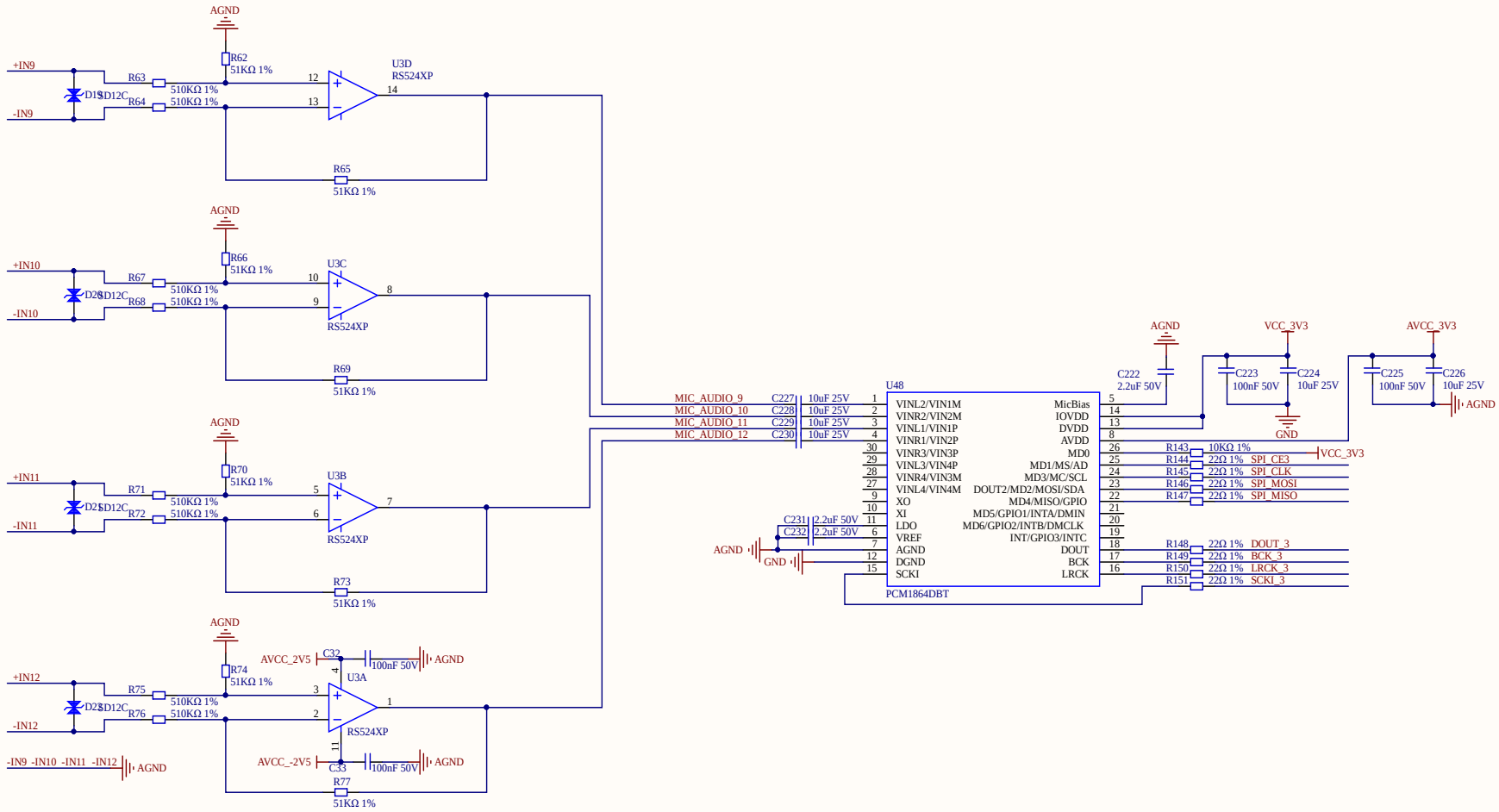

U20A
BANK1
IO DIFFIO_L3n DATA1,ASDO
IO VREFB1N0
IO DIFFIO_L4p FLASH_nCE,nCSO
IO DIFFIO_L6p
IO DIFFIO_L6n
IO DATA0
6 FPGA_ASDO
7
8 FPGA_nCSO
10 FPGA_RUN
11
13 FPGA_DATA
EP4CE22E22C8
U20B
BANK2
IO DIFFIO_L8p
IO DIFFIO_L11p
IO VREFB2N0
IO RUP1
IO RDN1
28
30
31 FPGA_TXD
32 FPGA_RXD
33 FPGA_nRST
EP4CE22E22C8
U20C
BANK3
IO DIFFIO_B2p
IO
IO PLL1_CLKOUTp
IO PLL1_CLKOUTn
IO VREFB3N0
IO DIFFIO_B9n
IO DIFFIO_B10n
IO DIFFIO_B11p
39
42
43
44 DOUT_3
46 BCK_3
49 LRCK_3
50 SCKI_3
51 DOUT_4
EP4CE22E22C8
U20D
BANK4
IO DIFFIO_B16p
IO DIFFIO_B17p
IO DIFFIO_B17n
IO
IO VREFB4N0
IO RUP2
IO RDN2
IO DIFFIO_B23n
IO DIFFIO_B24p
IO PLL4_CLKOUTp
IO PLL4_CLKOUTn
58 BCK_4
59 LRCK_4
60 SCKI_4
64 DOUT_5
65 BCK_5
66 LRCK_5
67 SCKI_5
68
69
71
72
EP4CE22E22C8
U20E
BANK5
IO RUP3
IO RDN3
IO VREFB5N0
IO DIFFIO_R11p
IO DIFFIO_R10p
IO DIFFIO_R9n DEV_OE
IO DIFFIO_R9p DEV_CLRn
76
77
80
83 DOUT_6
85 BCK_6
86 LRCK_6
87 SCKI_6
EP4CE22E22C8
U20F
BANK6
IO DIFFIO_R5n INIT_DONE
IO DIFFIO_R5p CRC_ERROR
IO
IO DIFFIO_R4n nCEO
IO DIFFIO_R4p CLKUSR
IO
IO VREFB6N0
IO DIFFIO_R1n PADD20
98
99
100
101
103 DOUT_7
104 BCK_7
105 LRCK_7
106 SCKI_7
EP4CE22E22C8
U20G
BANK7
IO DIFFIO_T23p
IO DIFFIO_T22p
IO PLL2_CLKOUTn
IO PLL2_CLKOUTp
IO RUP4
IO RDN4
IO VREFB7N0
IO DIFFIO_T19n PADD1
IO DIFFIO_T17p PADD4
IO DIFFIO_T13p PADD12
110
111 DOUT_8
112 BCK_8
113 LRCK_8
114 SCKI_8
115 R184 22Ω 1% FPGA_RST
119 R57 22Ω 1% DSP_IO11
120 R205 22Ω 1% MDRB
121 R206 22Ω 1% MFSRB
125 R207 22Ω 1% MCLKRB
EP4CE22E22C8
U20H
BANK8
IO DIFFIO_T10n DATA2
IO DIFFIO_T10p DATA3
IO DIFFIO_T9p DATA4
IO VREFB8N0
IO DATA5
IO DIFFIO_T2p
IO DATA12
IO PLL3_CLKOUTn
IO PLL3_CLKOUTp
132
133 DOUT_1
135 BCK_1
136 LRCK_1
137 SCKI_1
141 DOUT_2
142 BCK_2
143 LRCK_2
144 SCKI_2
EP4CE22E22C8
U20I
CLK1 DIFFCLK_0n
CLK2 DIFFCLK_1p
CLK3 DIFFCLK_1n
CLK4 DIFFCLK_2p
CLK5 DIFFCLK_2n
CLK6 DIFFCLK_3p
CLK7 DIFFCLK_3n
CLK8 DIFFCLK_5n
CLK9 DIFFCLK_5p
CLK10 DIFFCLK_4n
CLK11 DIFFCLK_4p
CLK12 DIFFCLK_7n
CLK13 DIFFCLK_7p
CLK14 DIFFCLK_6n
CLK15 DIFFCLK_6p
23 CLK_50Mhz
24
25
91
90
89
88
126
127
128
129
55
54
53
52
EP4CE22E22C8
U20J
VCCA1
VCCD_PLL1
VCCA2
VCCD_PLL2
VCCA3
VCCD_PLL3
VCCA4
VCCD_PLL4
GNDA1
GNDA2
GNDA3
GNDA4
35
37
107
109
3
1
75
73
36
108
2
74
VCC_2V5
VCC_1V2
GND
EP4CE22E22C8
U20K
TDO
TMS
TDI
TCK
MSEL0
MSEL1
MSEL2
nCE
nSTATUS
DCLK
nCONFIG
CONF_DONE
20 FPGA_TDO
18 FPGA_TMS
15 FPGA_TDI
16 FPGA_TCK
94
96
97
21 R212 10KΩ 1%
9 R214 10KΩ 1%
12 FPGA_DCLK
14 R215 10KΩ 1%
92 R216 10KΩ 1%
VCC_2V5
VCC_3V3
GND
EP4CE22E22C8
U20L
VCCIO1
VCCIO2
VCCIO3
VCCIO3
VCCIO4
VCCIO4
VCCIO5
VCCIO6
VCCIO7
VCCIO7
VCCIO8
VCCIO8
VCCINT
GND
17
26
40
47
56
62
81
93
117
122
130
139
5
29
34
38
45
61
70
78
84
102
116
124
134
138
0
4
19
22
27
41
48
57
63
79
82
95
118
123
131
140
VCC_3V3
VCC_1V2
GND
EP4CE22E22C8
U21
nCS
DATA
ASDI
DCLK
Vcc
Vcc
Vcc
GND
FPGA_nCSO
FPGA_DATA
FPGA_ASDO
FPGA_DCLK
22Ω 1% R204
1
2
5
6
8
7
3
4
C133 100nF 50V
EPCS16SI8N
VCC_3V3
GND
X2
VCC
OUT
EN/NC
GND
4
1
3
2
VCC_3V3
C134 100nF 50V
22Ω 1% CLK_50Mhz
R208
O9503250MEDA4SI
GND
VCC_2V5
C135 100nF 50V
C136 100nF 50V
C137 100nF 50V
C138 100nF 50V
C139 100nF 50V
GND
J2
GND
VCC_2V5
R210 1KΩ 1% FPGA_TCK 1
R211 10KΩ 1% FPGA_TDO 3
R213 10KΩ 1% FPGA_TMS 5
FPGA_TDI 9
1 2
3 4
5 6
7 8
9 10
IDC-10P
VCC_2V5
GND
VCC_3V3
R217 510Ω 1% FPGA_RUN
LED5 KP-2012SGC
VCC_3V3
R209 10KΩ 1%
FPGA_nRST
C140 100nF 50V
K2 SMD_3*6_4.3
GND
VCC_3V3
C141 100nF 50V
C142 100nF 50V
C143 100nF 50V
C144 100nF 50V
C145 100nF 50V
C146 100nF 50V
C147 100nF 50V
C148 100nF 50V
C149 100nF 50V
C150 100nF 50V
C151 100nF 50V
C152 100nF 50V
GND
VCC_1V2
C153 100nF 50V
C154 100nF 50V
C166 100nF 50V
C167 100nF 50V
C168 100nF 50V
C169 100nF 50V
C170 100nF 50V
C171 100nF 50V
C172 100nF 50V
C173 100nF 50V
C174 100nF 50V
C175 100nF 50V
C176 100nF 50V
GND
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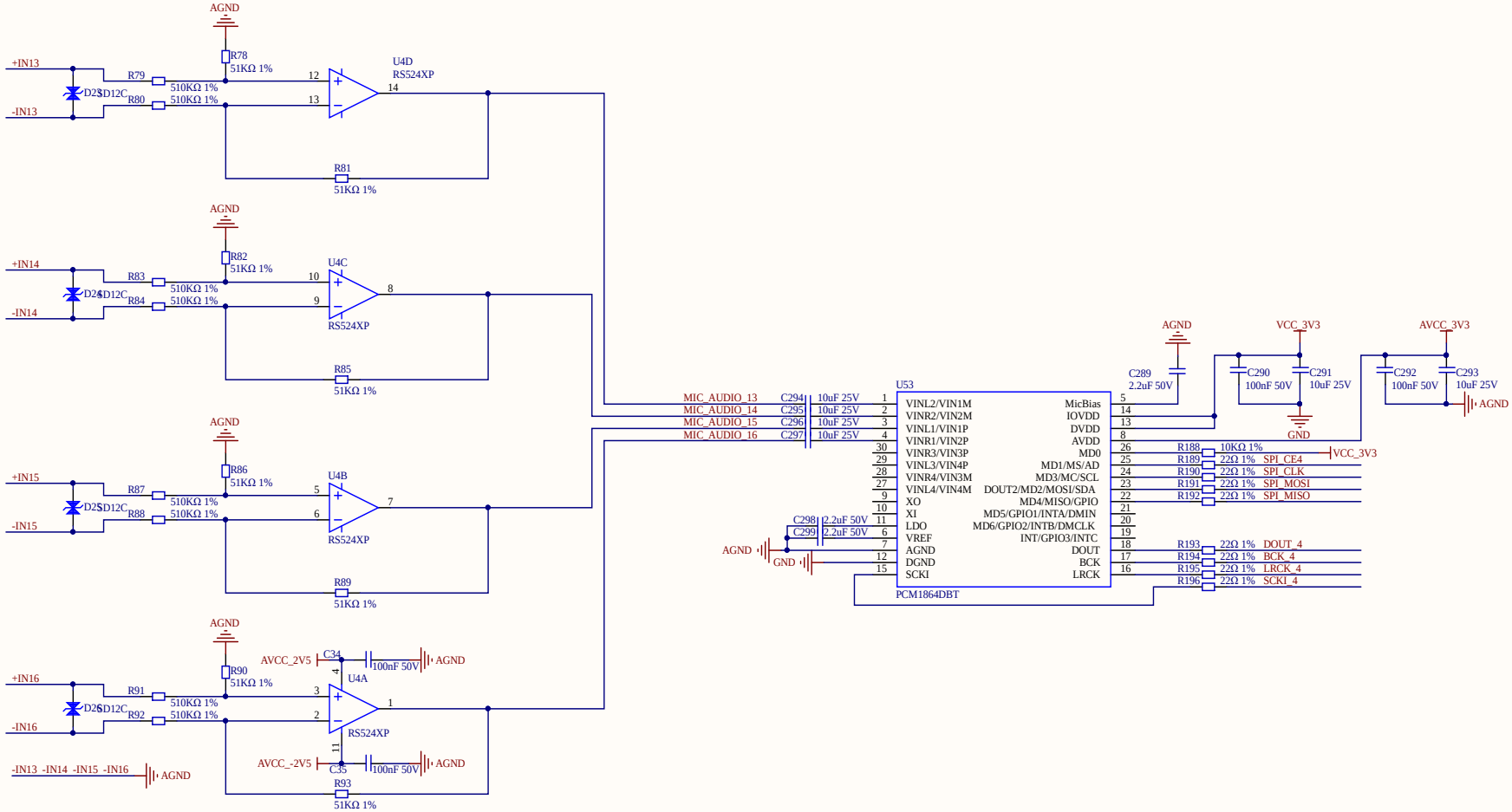
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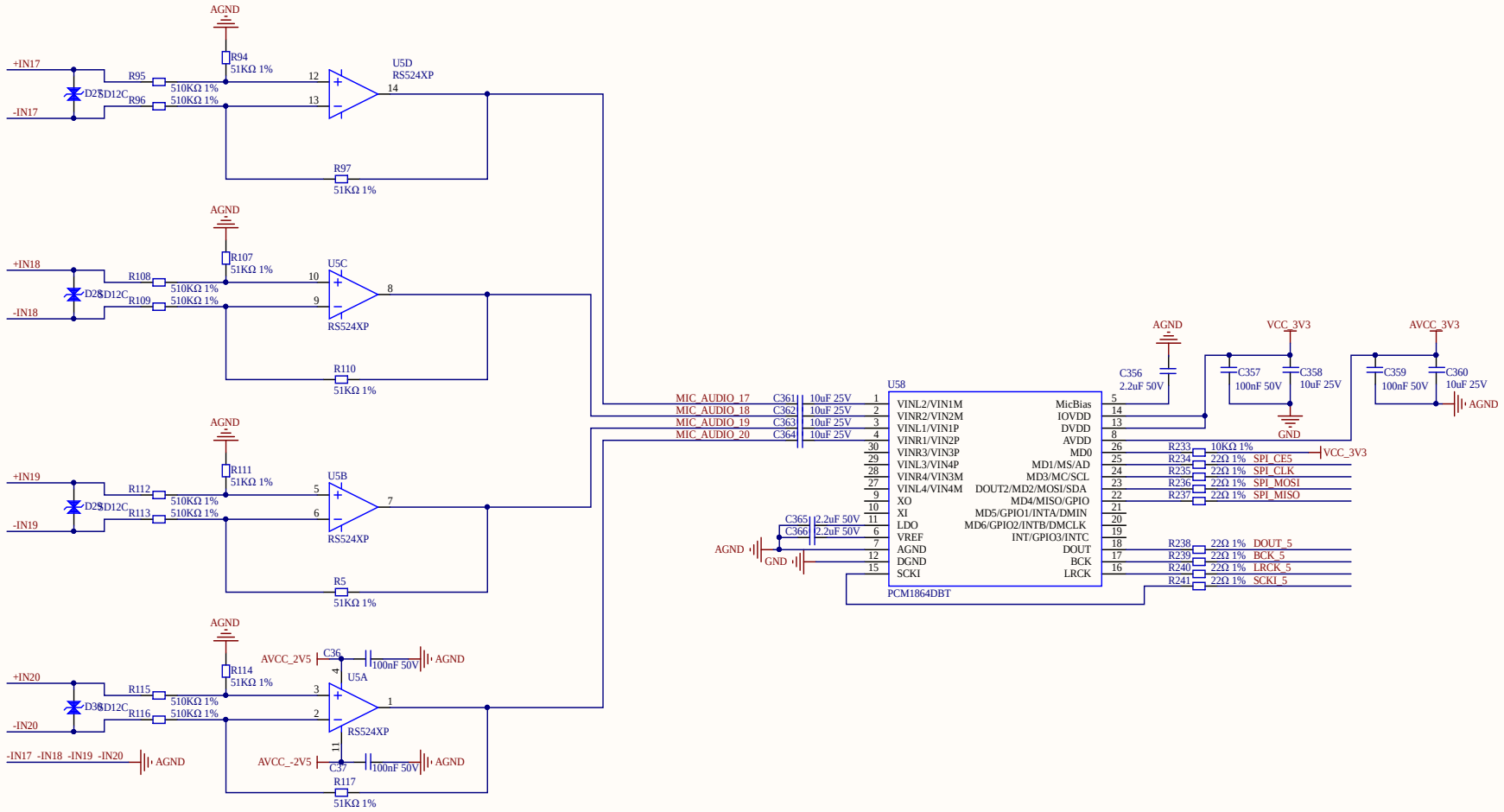
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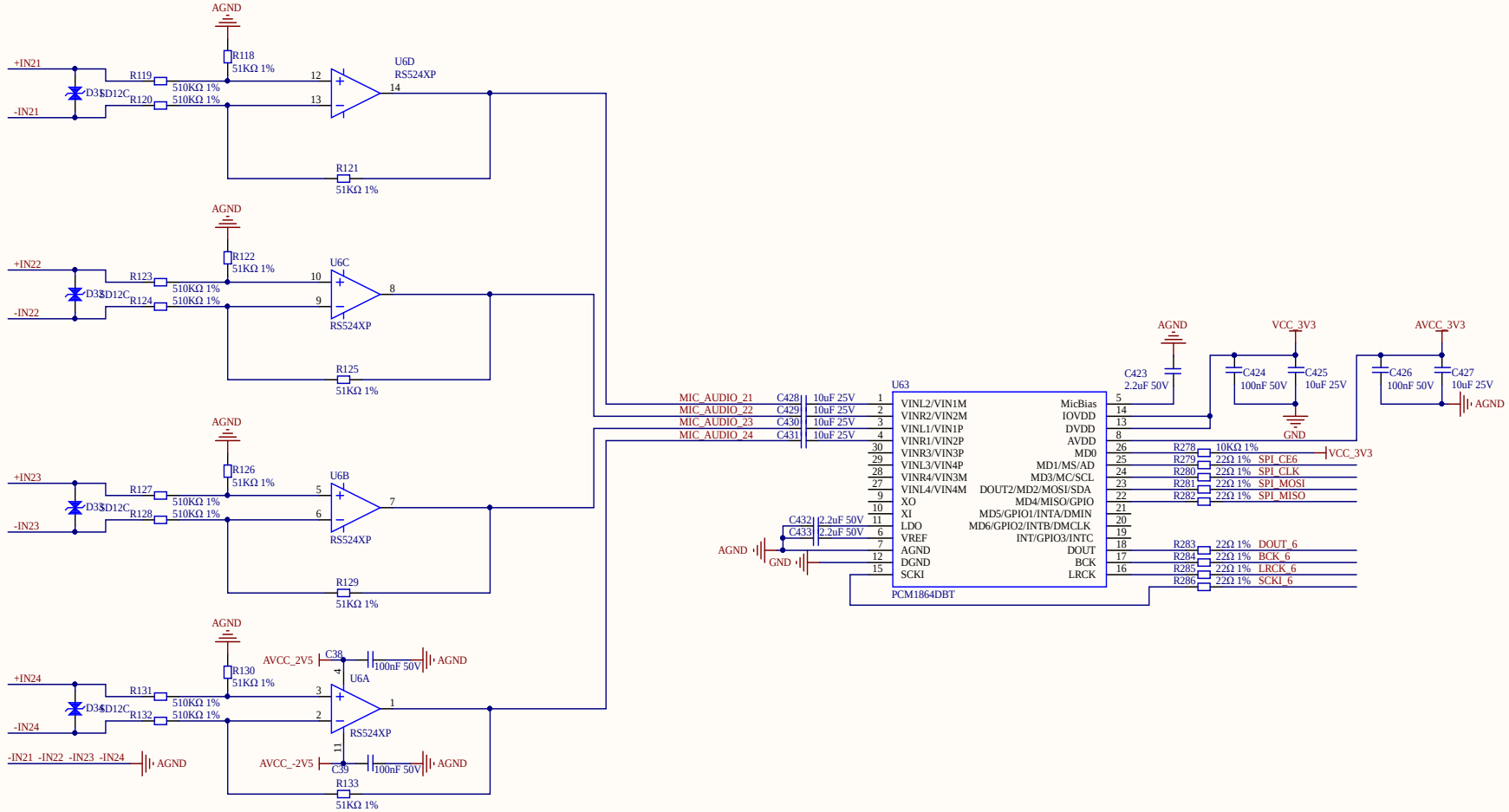
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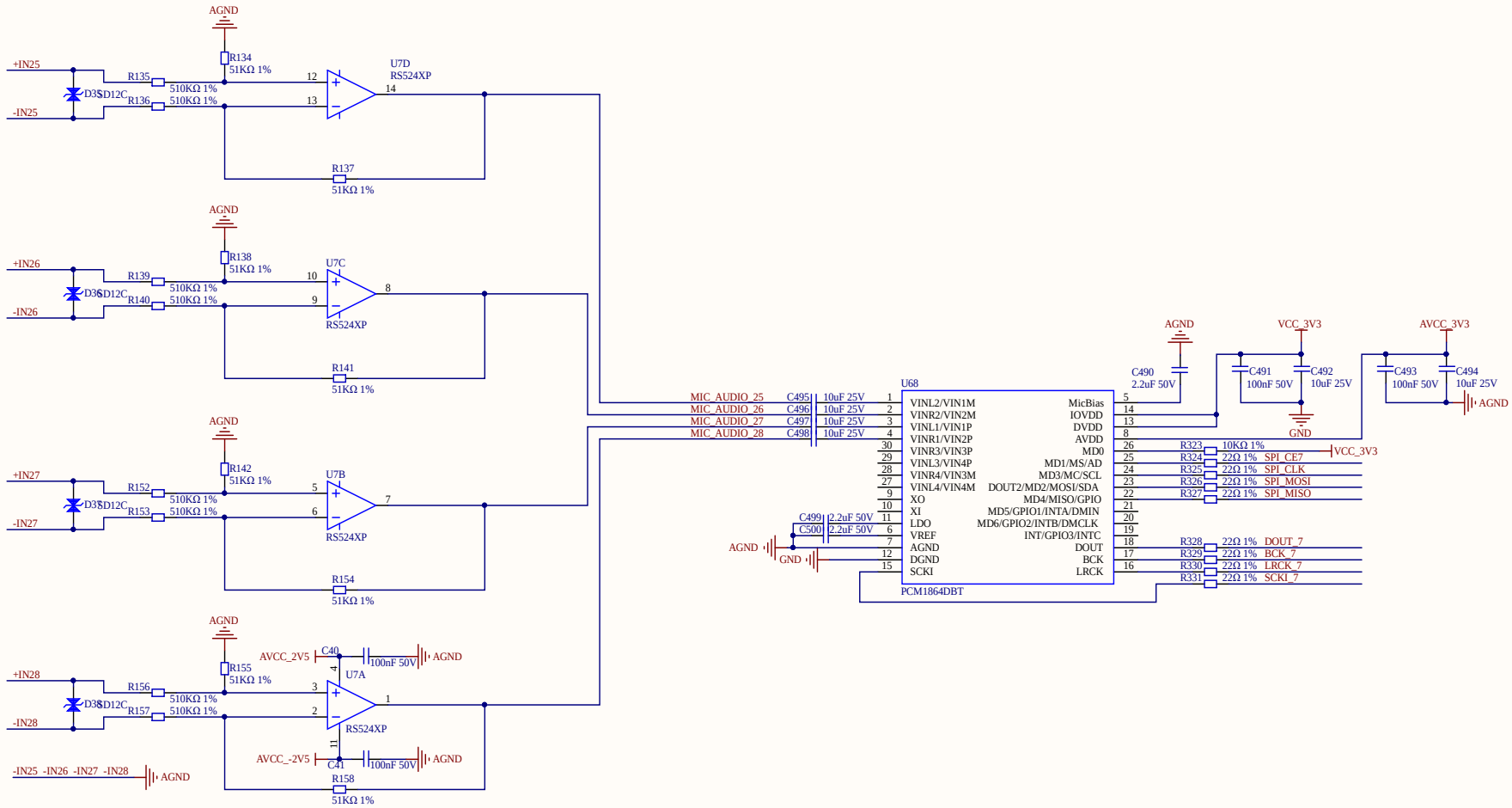
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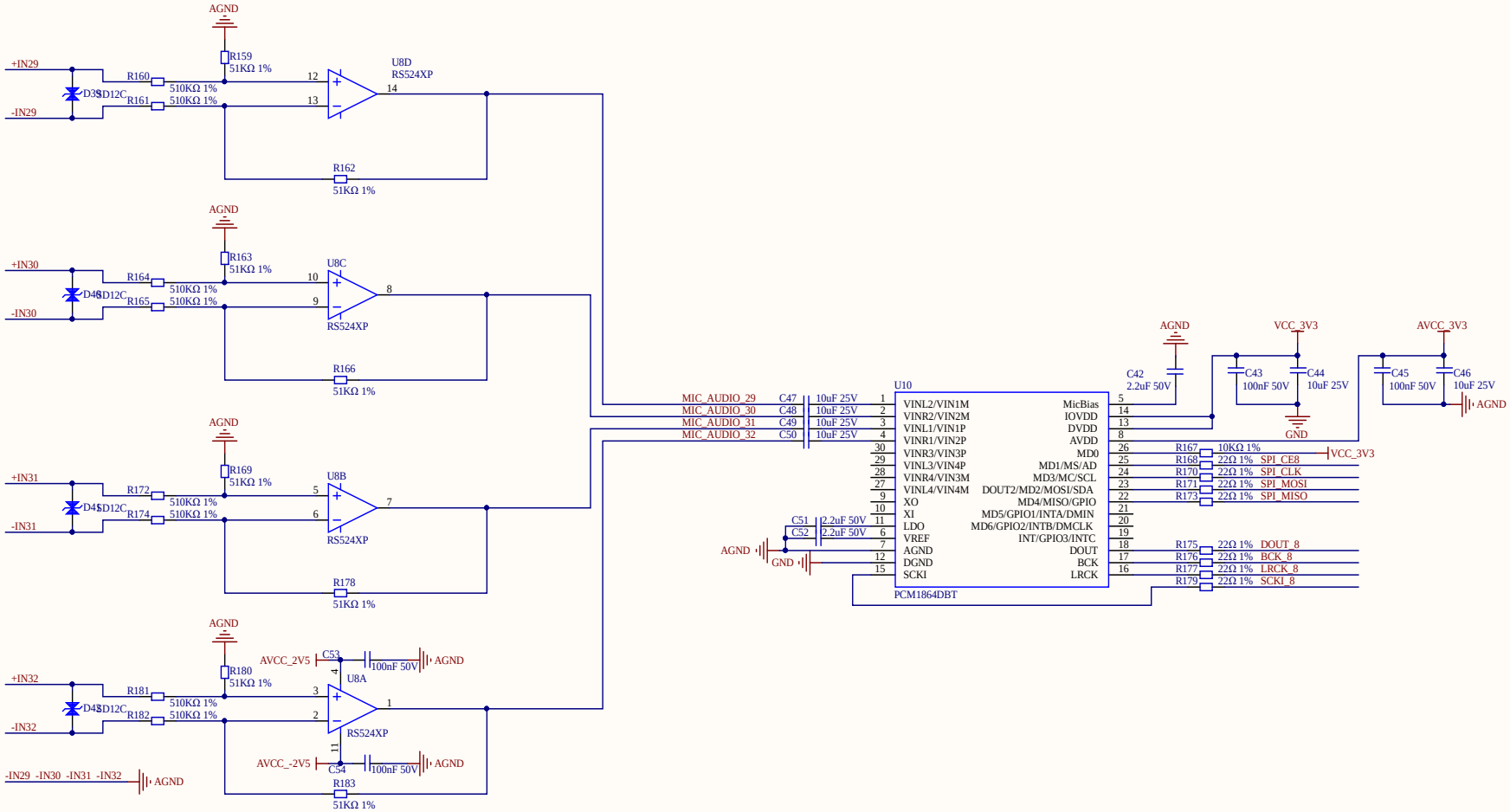
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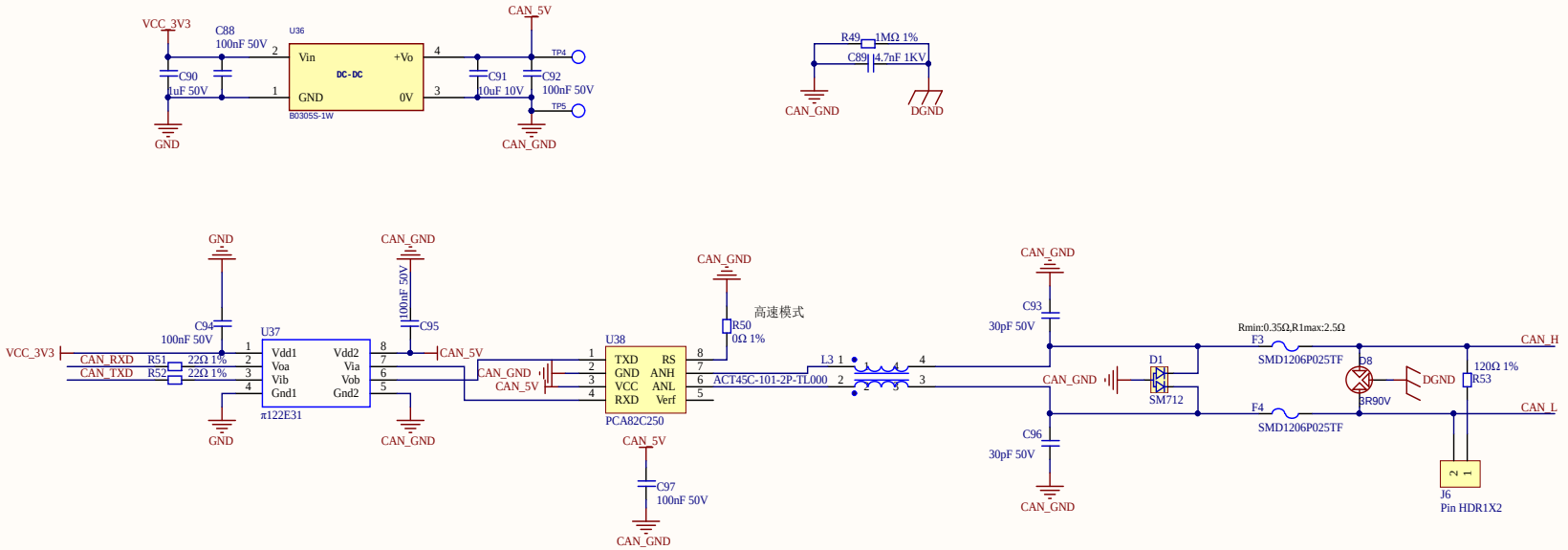
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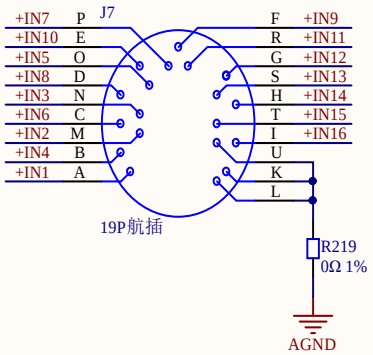


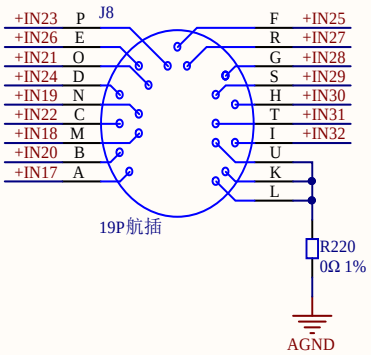
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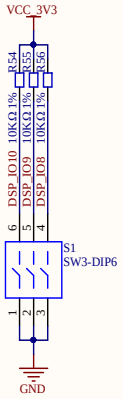
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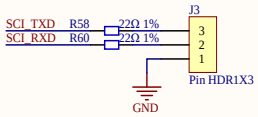


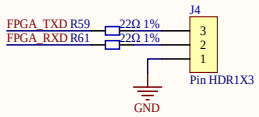




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嘉立创JLC7628（4层）

顺序 1：TOP Layer 4：Bottom Layer

